

Question:

There exist real numbers x and y , both greater than 1, such that $\log_x(y^x)=\log_y(x^{4y})=10$. Find xy .

Reasoning + Answer:

7 73 643 134" data-label="Text">

1. We are given the two logarithmic equations involving x and y :

$$\log_x(y^x)=10 \quad \text{and} \quad \log_y(x^{4y})=10$$
7 152 568 278" data-label="Text">

2. First, simplify $\log_x(y^x)$:

$$\log_x(y^x)=x \log_x(y) \quad \text{(by the power rule)}$$

Given that $\log_x(y^x)=10$, it follows:

$$x \log_x(y)=10 \quad \text{(1)}$$
7 296 616 422" data-label="Text">

3. Simplify $\log_y(x^{4y})$:

$$\log_y(x^{4y})=4y \log_y(x) \quad \text{(by the power rule)}$$

Given that $\log_y(x^{4y})=10$, we have:

$$4y \log_y(x)=10 \quad \text{(2)}$$
7 440 569 501" data-label="Text">

4. From equation (1), write $\log_x(y)$ as $\frac{10}{x}$:

$$\log_x(y)=\frac{10}{x}$$
7 519 741 580" data-label="Text">

5. Recall that $\log_x(y)=\frac{1}{\log_y(x)}$. So:

$$\frac{1}{\log_y(x)}=\frac{10}{x} \implies \log_y(x)=\frac{x}{10}$$
7 598 397 964" data-label="Text">

6. Plug into equation (2) which states:

$$4y \cdot \log_y(x)=10$$

Substitute $\log_y(x)=\frac{x}{10}$:

$$4y \cdot \left(\frac{x}{10}\right)=10$$

Simplify:

$$\frac{4xy}{10}=10$$
$$\frac{2xy}{5}=10$$

Multiply both sides by 5:

$$2xy=50$$

Divide both sides by 2:

$$xy=25$$

<answer>25</answer>